

Navraj Singh Gill

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EXPERIENCE

Code Driven Labs

May 2026 – Present

Software Engineering Intern

- Built **40 REST endpoints** across a **FastAPI** + **PostgreSQL** social-travel backend and full-stack admin dashboard, spanning 8 data domains backed by 8 **SQLAlchemy 2.0** models and 8 **Alembic** migrations (roughly 3,600 lines of Python)
- Shipped passwordless **email-OTP** + **JWT** authentication, **AWS S3** media uploads, and real-time **WebSocket** notifications powering a live activity feed
- Developed a **React / TypeScript** (Vite) admin frontend with 16 reusable components (about 1,700 lines), adding server-side search, sorting, filtering, and pagination over a seeded dataset of 500+ users

Wade Maritime Pvt. Ltd.

April – July 2024

Backend Intern

- Wrote and tuned **PostgreSQL queries** powering internal reporting dashboards, working under senior engineer review
- Built **Python FastAPI** endpoints supporting dynamic website features, with request validation handled through **Pydantic** models
- Resolved backend bug tickets and refactored existing modules inside a production codebase, participating in the team's code review workflow

Solutions 1313

May – Aug. 2022

Frontend Intern

- Built frontend components for client websites using **HTML, CSS, and JavaScript** alongside the development team
- Reworked existing JavaScript to improve cross-browser consistency and responsive behaviour across screen sizes
- Used **Chrome DevTools** to diagnose and resolve UI and layout issues

PROJECTS

2026 World Cup Predictor | *Python, XGBoost, scikit-learn, pandas, Monte Carlo*

Personal Project

- Engineered **rolling Elo ratings** over 49,000+ international matches dating to 1872 (updated on result and goal difference, weighted higher for World Cup fixtures) to derive team strength, which is not present in the source data
- Trained **XGBoost** against a logistic regression baseline on a **time-based split** (train pre-2024, test Jan 2024 – June 2026, $n = 2,543$): 59.97% vs 59.93% accuracy and **0.8689 vs 0.8706 log-loss**, keeping the baseline in the repo to verify the boosted model earned its complexity
- Logged **timestamped pre-kickoff predictions for 7 live knockout matches** and **Monte Carlo**-simulated the remaining bracket after each round; scored **7/7** under symmetrized win probability and 6/7 under raw per-row scoring, with the discrepancy documented rather than resolved in the model's favour

InSync | *Python, FastAPI, Groq LLM, Chrome API*

Personal Project

- Built a **Chrome extension** with a **FastAPI** backend that uses an LLM to categorize open browser tabs into color-coded native Chrome groups
- Added a caching layer keyed on tab metadata to cut repeated API calls, plus uncertainty-aware logic that flags low-confidence tabs instead of misfiling them

MolCraft | *Flutter, Figma, Google Play Store*

Personal Project

- Developed and published a **Flutter** mobile app on the **Google Play Store** that displays interactive 3D models of common chemistry molecules
- Designed the interface in **Figma**, translating 2D textbook diagrams into touch-controlled 3D structures aimed at students

Automatic Irrigation System | *Arduino, IoT Sensors*

IoT Innovation Challenge, IIT Bombay

- Built an **Arduino-based** setup with soil moisture, humidity, rain, and light sensors to automate watering for drought-prone test plots, with sensor-driven pump control and data logging
- Shortlisted among **15 teams nationally** to present at an IoT showcase hosted at **IIT Bombay**

Prompt Engineering in AI Models | *Independent Article*

Writing

- Tested **200 prompts across six prompting techniques** in a self-directed study on how prompting style affects response quality and where common failure modes appear

EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Computer Science (BCS), Candidate

2025 – 2030

- Relevant Coursework: Elementary Algorithm Design & Data Abstraction, Tools & Techniques for Software Development, Linear Algebra 1 (Honours), Calculus 2 (Honours)

Strawberry Fields High School

Class XII: 93.2%, Class X: 95.2%

Graduated 2025